Transductive Visual-Semantic Embedding for Zero-shot Learning

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ABSTRACT

Zero-shot learning (ZSL) aims to bridge the knowledge transfer via available semantic representations (e.g., attributes) between labeled source instances of seen classes and unlabelled target instances of unseen classes. Most existing ZSL approaches achieve this by learning a projection from the visual feature space to the semantic representation space based on the source instances, and directly applying it to the target instances. However, the intrinsic manifold structures residing in both semantic representations and visual features are not effectively incorporated into the learned projection function. Moreover, these methods may suffer from the inherent projection shift problem, due to the disjointness between seen and unseen classes. To overcome these drawbacks, we propose a novel framework termed transductive visual-semantic embedding (TVSE) for ZSL. In specific, TVSE first learns a latent embedding space to incorporate the manifold structures in both labeled source instances and unlabeled target instances under the transductive setting. In the learned space, each instance is viewed as a mixture of seen class scores. TVSE then effectively constructs the relational mapping between seen and unseen classes using the available semantic representations, and applies it to map the seen class scores of the target instances to their predictions of unseen classes. Extensive experiments on four benchmark datasets demonstrate that the proposed TVSE achieves competitive performance compared with the state-of-the-arts for zero-shot recognition and retrieval tasks.

CCS CONCEPTS

• **Computing methodologies** → **Transfer learning**; *Supervised learning by classification*;

KEYWORDS

Zero-shot learning; transductive learning; matrix factorization; manifold learning

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1 INTRODUCTION

Conventional visual recognition systems usually require a lot of manually labeled training data to achieve good classification accuracy, typically several thousand images for each class to be learned. Due to the ever increasing number of available images and categories to be recognized, it becomes infeasible to label images for each possible class. In the real-world setting, it would be desirable for visual recognition systems to be able to recognize instances of those rare classes, while demanding minimum human efforts and labeled examples. Zero-shot learning (ZSL) [13, 15] has long been believed to hold the key to the above problem of

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recognition a new object class without ever seeing a visual instance.

In ZSL, two sets of instances with disjoint classes are considered: labeled source instances of seen classes, and unlabeled target instances to be classified in new classes that have never been seen before. ZSL requires to extract the knowledge from the source set and to transfer that knowledge to the target set. This is often achieved by mapping both seen and unseen classes to a semantic embedding space where each class label/name is represented as a high dimensional vector. A commonly adopted semantic representation space is constructed based on the available semantic representations of classes including binary vectors of visual attributes [13, 16] and continuous word vectors [7, 20] encoding linguistic context. For example, the class "polar bear" can be represented as a binary attribute vector indicating that it is "while", "has fur" and "eat fish". Given a semantic embedding space, most existing approaches [1, 14, 27] for ZSL first learns a projection function that maps the low-level visual features to the semantic representations of seen classes from the source set by traditional classification or regression models. The learned projection function is then applied to the target set to map each target instance into the same embedding space. Finally, the classification of these target instances can be obtained by distance/similarity matching to the semantic representations of unseen classes.

However, two inherent problems exist in these projection based ZSL approaches. The first one is that the intrinsic manifold structure residing in the available semantic representations of classes is not well explored when learning the projection function. Indeed, the semantically related classes generally lie in the same manifold and the manifold structure is rational to be associated with the visual features of the corresponding classes. The second problem is the inherent projection shift problem, which has also been addressed in [9, 30]. The source and target sets may have different data distribution due to the disjoint classes that are semantically unrelated. Therefore, using the projection function learned between the visual features and the semantic representations of seen classes in the source set, are prone to unknown bias when predicting unseen classes in the target set.

In this paper, we propose to tackle the aforementioned problems by developing a novel framework termed Transductive Visual-Semantic Embedding (TVSE). The flowchart of the proposed TVSE is illustrated in Figure 1. Different from the works [10, 30] that directly model the manifold structures based on the available semantic representations of classes, TVSE turns to learning a common latent embedding space, where each instance of both source and target sets is viewed as a mixture of seen classes scores. The latent embedding space is learned by performing collective matrix factorization with graph regularization on both source and target instances under the transductive setting, ensuring the coherent manifold structures of visual features and classes are effectively incorporated for knowledge transfer. For example, the instances in the same class form a cluster in the embedding space, and the clusters of the semantically related classes (e.g, "Lazuli_Bunting" and "Painted_Bunting") are nearer in the space. Furthermore, to overcome the issue of inherent projection shift, an effective scheme is developed for TVSE to construct the relational mapping between seen and unseen classes based on the available semantic representations of classes. For example, the relation between a seen class (e.g., "Lazuli_Bunting") and all the unseen classes can be obtained based on the relational mapping. Finally, the seen class proportions of the target instances can be mapped to their predictions of unseen classes.

The main contributions of our work are three-folds: 1) We propose a novel framework termed TVSE for zero-shot learning. TVSE utilizes collective matrix factorization technique with graph regularization to learn a latent embedding space, where the valuable manifold structures of both visual features and classes are effectively modeled. 2) We also develop a scheme in TVSE to effectively construct the relational mapping between the seen and unseen classes. 3) TVSE can overcome the inherent drawbacks in previous projection based ZSL approaches. We conduct extensive experiments on four widely used benchmark datasets on zero-shot recognition task, showing that TVSE achieves competitive performance compared with the state-of-the-arts for zero-shot recognition and retrieval tasks.

The remainder of the paper is organized as follows. In the next section, we briefly review the related methods for ZSL. Then, we introduce our proposed framework, followed by experimental results with comprehensive analysis on four benchmark datasets. Finally, we draw the conclusion.

2 RELATED WORK

Recently, ZSL has received a growing amount of attention [8, 11, 15, 18, 22, 31, 32, 35]. Existing ZSL methods differ in two aspects: 1) the available semantic representations used and 2) how the knowledge is transferred from the seen classes to the unseen classes.

The available semantic representations of classes typically include visual attributes and semantic word vectors. As an appealing source of information, attributes (binary or continuous) describe common properties of object classes and can be acquired from domain experts or crowd-sourcing techniques. However, the attributes need to be manually annotated, which is of high cost and limited to large-scale recognition. As an alternative to visual attributes, the methods based on semantic word vectors have gained popularity. In this case, a word space is extracted from linguistic knowledge bases, e.g., Wikipedia by independent natural language processing model [19]. Then each class's textual name can be represented by a continuous vector using the language model. Importantly, regardless of the available semantic representations used, most existing methods focus on how to best learn the projections. However, they may fail to explore the manifold structures residing in the semantic representations of classes and to associate these valuable structures to the visual features of the

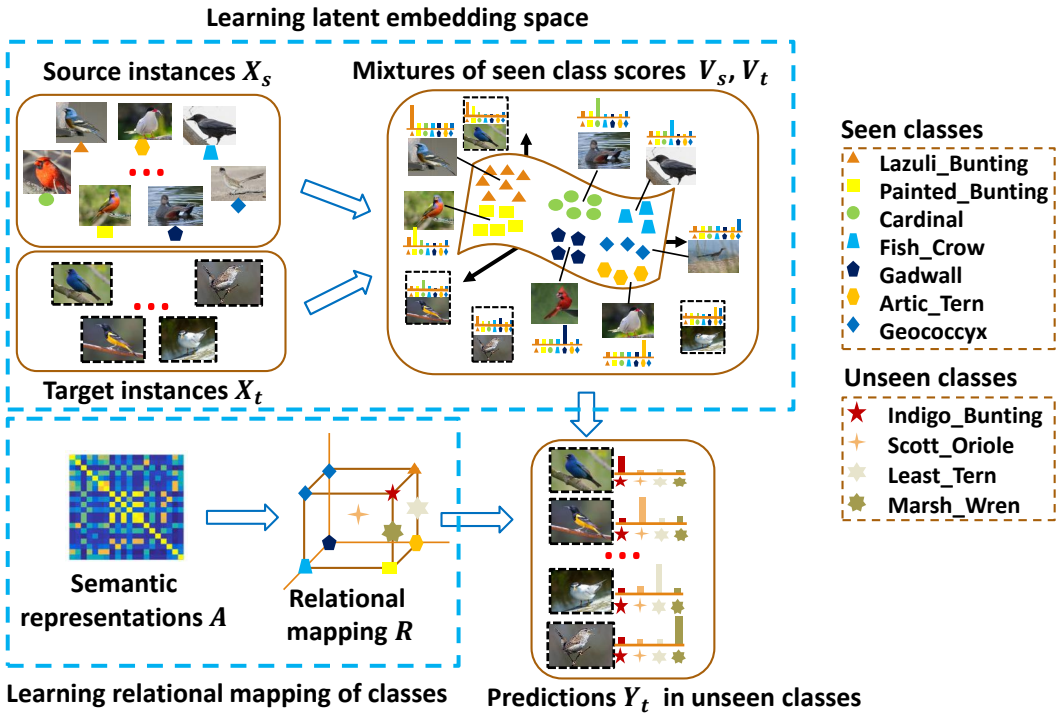


Figure 1: The general flowchart of the proposed TVSE on CUB [29] dataset. TVSE represents each instance in both source and target set are represented by a mixture of seen class scores by learning a latent embedding space under the transductive setting. It then maps the seen class scores of the target instances (in black dot rectangles) to their predictions of unseen classes via the relational mapping built from the available semantic representations.

instances in both source and target sets. In our work, we propose to learn a latent embedding space under the transductive setting to explore the manifold structures of both source and target instances. In addition, different from previous transductive ZSL approaches [9, 24] that employ generative model or graph-based label propagation scheme, the manifold structures in both source and target instances are explored via an advanced technique of collective matrix factorization with graph regularization.

To bridge the knowledge transfer between seen and unseen classes, according to the discussion in [10], existing ZSL methods can be grouped into two categories: classification based and projection based. The projection based methods [1, 7, 31, 34] directly embed the visual feature vectors into a semantic embedding space defined by the available semantic representations via learning a projection function. Such a function is learned from the labeled source instances containing seen classes only. By applying the same visual feature embedding process on the target instances, the predictions of unseen classes can be obtained directly by nearest neighbor matching or its variants [14]. In contrast, the classification based methods [13, 21] rely on learning a n-way discrete classifier for the seen classes using the visual features of the instances in the source set. The classifier is then used to compute the visual similarity between an instance of unseen classes in the target set to those of seen classes in the source set. The semantic relatedness between the seen and unseen classes is modeled by the distance between their available semantic representations. Indeed, our work has the characteristics of both categories. Specifically, compared with the projection based methods, we learn a latent embedding space which directly associates the relations between visual features and seen classes. Similar to the classification based ones, in our work, each instance in both source and target sets can be viewed as a mixture of seen classes scores, and we also model the semantic relatedness between seen and unseen classes by weighted graph, rather than the distance matching.

3 TRANSDUCTIVE VISUAL-SEMANTIC EMBEDDING

3.1 Problem Statement

Suppose we have c_s seen classes with n_s labeled instances in source set $D_s = \{\mathbf{X}_s, \mathbf{Y}_s\}$, and c_t unseen classes with n_t unlabeled instances in target set $D_t = \{\mathbf{X}_t, \mathbf{Y}_t\}$. The indices of the source and target sets are $\mathcal{I} = \{s, t\}$. Let $\mathbf{X}_\pi = [\mathbf{x}^\pi_{*1}, ..., \mathbf{x}^\pi_{*n_\pi}] \in \mathbb{R}^{m \times n_\pi}$ denote the m-dimensional visual feature vectors of source and target instances with $\mathbf{x}^\pi_{*i}$ being the ith instances in $\mathcal{D}_\pi$, $\pi \in \mathcal{I}$. Class labels of the instances

in $\mathcal{D}_\pi$ are denoted as $\mathbf{Y}_\pi = [\mathbf{y}^\pi_{1*}, ..., \mathbf{y}^\pi_{n_\pi *}] \in \mathbb{R}^{c_\pi \times n_\pi}$, where $\mathbf{y}^\pi_{i*}$ is the one-hot vector for the ith instance in D_π. Note that the instances in D_s and D_t have disjoint class labels, i.e. $\mathbf{Y}_s \cap \mathbf{Y}_t = \emptyset$. In addition, the available semantic representations (e.g., attributes) of classes in D_π are denoted as $\mathbf{A}_\pi = [\mathbf{a}_1, ..., \mathbf{a}_{n_\pi}] \in \mathbb{R}^{d \times n_\pi}$, where $\mathbf{a}_i$ represents an d-dimensional binary/continuous vector for the ith class. The problem of zero-shot learning (ZSL) is to estimate $\mathbf{Y}_t$ given $\mathbf{X}_t$ and $\mathbf{A}_t$.

To this end, our proposed *Transductive Visual-Semantic Embedding* (TVSE) is to learn a common latent embedding space underlying both source and target sets as the bridge for smooth knowledge transfer. One one hand, the learned latent embedding space is expected to overcome the limitations of the available semantic representations to incorporate the intrinsic manifold structures (visual features and classes) that are coherent in both source and target sets. One the other hand, it is also desired to tackle the problem of the inherent projection shift during knowledge transfer from the source instances to the target instances.

3.2 The Proposed Framework

To achieve the goal of TVSE, we propose an algorithm that utilizes collective matrix factorization technique combining with graph regularization under the transductive setting. We then construct the relational mapping between seen and unseen classes through the weight graph method.

3.2.1 Learning the latent embedding space. We assume that a latent embedding space connects the visual features and the seen classes in the source set $\mathcal{D}_s$. Besides, under the transductive setting, the target set $\mathcal{D}_t$ is also taken into account during learning the latent embedding space. Thus, the connection between the visual features in $\mathcal{D}_t$ and the seen classes in $\mathcal{D}_s$ is also constructed. To ensure smooth knowledge transfer from $\mathcal{D}_s$ to $\mathcal{D}_t$, the latent embedding space is required to be shared between the two sets, and each instance in each set can be represented by a mixture of latent factors in the embedding space. These latent factors can be obtained by applying collective matrix factorization [17] on the visual feature matrices $\mathbf{X}_s$ and $\mathbf{X}_t$ jointly, to minimize the reconstruction loss

$$\begin{aligned}\mathcal{L} &= \|\mathbf{X}_s - \mathbf{U}_s\mathbf{H}\mathbf{V}_s^\top\|^2 + \|\mathbf{X}_t - \mathbf{U}_t\mathbf{H}\mathbf{V}_t^\top\|^2, \\ &= \sum_{\pi \in \mathcal{I}} \|\mathbf{X}_\pi - \mathbf{U}_\pi\mathbf{H}\mathbf{V}_\pi^\top\|^2, \end{aligned} \tag{1}$$

where $\|\cdot\|^2$ is the Frobenius norm of a matrix. In Eq. 1, $\mathbf{U}_\pi = [\mathbf{u}^\pi_{*1}, ..., \mathbf{u}^\pi_{*k}] \in \mathbb{R}^{m \times k}$ with each $\mathbf{u}_{*i}$ representing a visual feature cluster. $\mathbf{V}_\pi = [\mathbf{v}^\pi_{*1}, ..., \mathbf{v}^\pi_{*c_s}] \in \mathbb{R}^{n_s \times c_s}$ with each $\mathbf{v}_i$ being the mixture of seen class scores. $\mathbf{H} \in \mathbb{R}^{k \times c_s}$ denotes the latent factors residing in the seen classes, and it also builds the associations between the feature clusters $\mathbf{U}_\pi$ of each set $\mathcal{D}_\pi$ and their mixtures of seen class scores $\mathbf{V}_\pi$. Different from the previous work [25] that considers the available semantic representations $\mathbf{A}_s$ of seen classes explicitly as the latent factors, here $\mathbf{H}$ is learned from both the source instances of seen classes and the unlabeled target instances. Therefore, $\mathbf{H}$ is more effective to incorporate informative associations between visual features in each set and the seen classes.

Given the seen class information $\mathbf{Y}_s$ for instances in $\mathcal{D}_s$, we enforce $\mathbf{V}_s \equiv \mathbf{Y}_s$ as in [17] to make the latent embedding space effectively bridge the connection between the visual features and seen classes. Through the connection, the instances in $\mathcal{D}_t$ can also be appropriately represented by mixtures of seen class scores ($\mathbf{V}_t$) in the latent embedding space, with the knowledge from $\mathbf{Y}_s$ in $\mathcal{D}_s$ being smoothly transferred to the $\mathbf{V}_t$ in $\mathcal{D}_s$.

3.2.2 Modeling the manifold structures of instances. For ZSL, it is expected that the geometric structures in the labeled source instances can be preserved for smooth transfer procedure. Specifically, if two instances $\mathbf{x}_i, \mathbf{x}_j$ in each dataset D_π are close the visual feature space, then their class label should also be close. This assumption can be reflected in their mixtures of seen class scores $\mathbf{v}_i$, $\mathbf{v}_j$ in the latent embedding space. Inspired by [4], the geometric structure can be modeled by a nearest neighbor graph in the instance space. Consider an instance graph G^I_π with n_I vertices with each representing an instance in set D_π. Then the pairwise similarity matrix can be defined as

$$(\mathbf{W}^I_\pi)_{ij} = \begin{cases} cos(\mathbf{x}^\pi_{*i}, \mathbf{x}^\pi_{*j}), & if\ \mathbf{x}^\pi_{*i} \in N_p(x^\pi_{*j})\ or\ \mathbf{x}^\pi_{*j} \in N_p(x^\pi_{*i}) \\ 0, & otherwise \end{cases} \tag{2}$$

where $N_p(x)$ denotes the top-p nearest neighbors of the ith instance $\mathbf{x}^\pi_{*i}$. By defining $\mathbf{Q}^I_\pi = diag(\sum_i (\mathbf{W}^I_\pi)_{ij})$, then preserving the visual feature manifold structure in D_π is derived to minimize the instance graph regularizer

$$R^I_\pi = \frac{1}{2}\sum_{ij} \|\mathbf{v}_i - \mathbf{v}_j\|^2 (\mathbf{W}^I_\pi)_{ij} = tr(\mathbf{V}_\pi^\top(\mathbf{Q}^I_\pi - \mathbf{W}^I_\pi)\mathbf{V}_\pi) \tag{3}$$

Furthermore, from the aspect of features, each dimension of the features is sampled from a distribution supported by another low-dimensional manifold. Therefore, we construct a feature graph G^F_π with n_F vertices each representing a feature in set D_π as

$$(\mathbf{W}^F_\pi)_{ij} = \begin{cases} cos(\mathbf{x}^\pi_{i*}, \mathbf{x}^\pi_{j*}), & if\ \mathbf{x}^\pi_{i*} \in N_p(\mathbf{x}^\pi_{j*})\ or\ \mathbf{x}^\pi_{j*} \in N_p(\mathbf{x}^\pi_{i*}) \\ 0, & otherwise, \end{cases} \tag{4}$$

where $N_p(\mathbf{x}^\pi_{i*})$ denotes the top-p nearest neighbors of the ith dimension of feature in $\mathbf{x}^\pi_{i*}$. Let $\mathbf{Q}^F_\pi = diag(\sum_i (\mathbf{W}^F_\pi)_{ij})$, preserving the geometric structure in each dataset D_π further includes minimizing the feature graph regularizer

$$R^F_\pi = \frac{1}{2}\sum_{ij} \|\mathbf{u}_i - \mathbf{u}_j\|^2 (\mathbf{W}^F_\pi)_{ij} = tr(\mathbf{U}_\pi^\top(\mathbf{Q}^F_\pi - \mathbf{W}^F_\pi)\mathbf{U}_\pi) \tag{5}$$

Clearly, the graph regularizer defined in Eq. 3 and Eq. 5 can preserve the coherent manifold structures for each dataset during the latent embedding space learning procedure. Therefore, we can incorporate the two regularization

terms into Eq. 1 to reach the final optimization problem as

$$\min_{\mathbf{U}_\pi,\mathbf{H},\mathbf{V}_\pi \geq 0} \mathcal{O} = \mathcal{L} + \sum_{\pi\in\mathcal{I}} (\lambda R_\pi^I + \mu R_\pi^F),$$
$$s.t.\ \mathbf{U}_\pi^\top \mathbf{1}_m = \mathbf{1}_k,\ \mathbf{V}_\pi^\top \mathbf{1}_{n_\pi} = \mathbf{1}_{c_s},\ \forall \pi \in \mathcal{I} \tag{6}$$

where λ, μ are regularization coefficients, and $\mathbf{1}_\Delta$, $\Delta \in \{m, k, c_s\}$ is the vector of ones. The ℓ_1 normalization constraints on each column of $\mathbf{U}_s$ and $\mathbf{V}_s$ are used to make the optimization well-defined. With the optimization results, for instances in the target set $\mathcal{D}_t$, their mixtures of seen classes scores $\mathbf{V}_t$ can be obtained.

Next, we discuss how to map $\mathbf{V}_t$ of the target instances to their predictions of unseen classes, which relies on the mapping between the seen and unseen classes.

3.2.3 Constructing the relational mapping of classes. Inspired by [5, 33, 34], we assume that each seen class can be represented as a combination of all unseen classes based on their semantic representations $\mathbf{A}_\pi = [\mathbf{a}_1, ..., \mathbf{a}_{n_\pi}]$. In order to effectively incorporate the intra-class relationship, we further transform the semantic representation of each class via kernel mapping, e.g., the simple linear kernel, where $\mathbf{\Phi}_\pi = [\boldsymbol{\phi}_1, ..., \boldsymbol{\phi}_{c_\pi}] \in \mathbb{R}^{c_\pi \times c_s}$ with $\boldsymbol{\phi}_i$ denoting the kernelized version of $\mathbf{a}_i$.

Given each $\boldsymbol{\phi}_i$ of a seen class, we propose a parameterized optimization problem inspired by the weight graph method [5], which minimizing the reconstruction error of each $\boldsymbol{\phi}_i$ based on the kernelized semantic representations of all unseen classes as follows:

$$\min \gamma \|\mathbf{R}\|^2 + \sum_{i=1}^{c_s} \|\boldsymbol{\phi}_s^i - \sum_{j=1}^{c_t} \boldsymbol{\phi}_t^j r_{ij}\|^2, \tag{7}$$

where $\gamma \geq 0$ is the regularization parameter, $\|\cdot\|^2$ denotes the ℓ_2 norm of a vector, and $\mathbf{R} = \{r_{ij}\} \in \mathbb{R}^{c_s \times c_t}$ describes the relational mapping matrix that contains the pairwise weight value for pairwise classes. Eq. 7 can be solved using standard quadratic programming as in [5].

3.2.4 Prediction for Target Dataset. For an instance $\mathbf{x}_{*,i}^t$ in the target dataset $\mathcal{D}_t$, given its mixture of seen class scores $\mathbf{v}_{i*}^t$, we can infer the prediction for $\mathbf{x}_{*,i}^t$ by

$$\mathbf{y}_{*,i}^t = argmax_j (\mathbf{v}_{i*}^t \mathbf{R})_j, \tag{8}$$

where $\mathbf{v}_{i*}^t R$ indicates the mixture of unseen class scores for $\mathbf{x}_{*,i}^t$ through the relational mapping $\mathbf{R}$ from the mixture of seen classes $\mathbf{v}_{i*}^t$. The operation $argmax(\cdot)$ selects the unseen class with the highest score as the prediction.

3.3 Optimization

We now discuss the solution to the optimization problem in Eq. 1. As the original formulation in Eq. 1 belongs to the constrained optimization problem, we add Lagrange functions for the parameters $\mathbf{U}_\pi$ and $\mathbf{V}_\pi$ with nonnegative and L1 normalization constraints to Eq. 1 as

$$\hat{\mathcal{L}} = \mathcal{O} + \sum_{\pi\in\mathcal{I}} tr(\Omega_\pi(\mathbf{U}_\pi^\top \mathbf{1}_m - \mathbf{1}_k)(\mathbf{U}_\pi^\top \mathbf{1}_m - \mathbf{1}_k)^\top)$$
$$+ \sum_{\pi\in\mathcal{I}} tr(\Theta_\pi(\mathbf{V}_\pi^\top \mathbf{1}_{n_\pi} - \mathbf{1}_{c_s})(\mathbf{V}_\pi^\top \mathbf{1}_{n_\pi} - \mathbf{1}_{c_s})^\top), \tag{9}$$

where $\Omega_\pi \in \mathbb{R}^{k\times k}$, $\Theta_\pi \in \mathbb{R}^{c\times c}$ are the Lagrange multipliers for the two constraints with respect to $\mathbf{U}_\pi$ and $\mathbf{V}_\pi$. We iteratively optimize one variable and derive its updating rule while fixing the remaining ones. The procedure will monotonically decrease the objective function in Eq. 9 until convergence. Similar as the derivations in [17], the updating rules for each variable in $\{\mathbf{U}_\pi, \mathbf{V}_\pi\}$ can be achieved by using the Karush-Kuhn-Tucker (KKT) complementarity condition [3] and setting their derivations to zero, which leads to the following updating formula respectively:

$$\mathbf{U}_\pi \leftarrow \mathbf{U}_\pi \circ \sqrt{\frac{\mathbf{X}_\pi \mathbf{V}_\pi \mathbf{H}^\top + \lambda \mathbf{W}_\pi^I \mathbf{U}_\pi}{\mathbf{U}_\pi \mathbf{H} \mathbf{V}_\pi^\top \mathbf{V}_\pi \mathbf{H}^\top + \lambda \mathbf{Q}_\pi^I \mathbf{U}_\pi}}, \tag{10}$$

$$\mathbf{V}_\pi \leftarrow \mathbf{V}_\pi \circ \sqrt{\frac{\mathbf{X}_\pi^\top \mathbf{U}_\pi \mathbf{H} + \mu \mathbf{W}_\pi^F \mathbf{V}_\pi}{\mathbf{V}_\pi \mathbf{H}^\top \mathbf{U}_\pi^\top \mathbf{U}_\pi \mathbf{H} + \mu \mathbf{Q}_\pi^F \mathbf{V}_\pi}}, \tag{11}$$

where $\circ$ denotes the element-wise operations for the matrix computation. For variable H without constraints, we can also derive the close-form updating rule for it, as:

$$\mathbf{H} \leftarrow \mathbf{H} \circ \sqrt{\frac{\sum_{\pi\in\mathcal{I}} \mathbf{U}_\pi^\top \mathbf{X}_\pi \mathbf{V}_\pi}{\sum_{\pi\in\mathcal{I}} \mathbf{U}_\pi^\top \mathbf{U}_\pi \mathbf{H} \mathbf{V}_\pi^\top \mathbf{V}_\pi}}. \tag{12}$$

Algorithm 1 TVSE for Zero Shot Learning.

Input: Matrices $\mathbf{X}_\pi$, $\mathbf{A}_\pi$ and $\mathbf{Y}_s$, parameters k, p, λ, γ;
Output: Predictions $\mathbf{Y}_t$ for instances in target dataset $\mathcal{D}_t$.

1: Normalize $\mathbf{X}_\pi$ with ℓ_2 normalization and build graphs G_π^I and G_π^F.
2: Initialize $\mathbf{U}_\pi$, $\mathbf{H}$ as random positive matrix respectively, $\mathbf{V}_s$ as $\mathbf{Y}_s$, $\mathbf{V}_t$ by ridge regression trained on $\{\mathbf{X}_s, \mathbf{Y}_s\}$.
3: **repeat**
4: Update $\mathbf{U}_\pi$, $\mathbf{V}_\pi$ and $\mathbf{H}$ by Eq. 10 $\sim$ Eq. 12.
5: Set $\mathbf{V}_s = \mathbf{Y}_s$ and normalize each column of $\mathbf{U}_\pi$ and $\mathbf{V}_\pi$ by ℓ_1 normalization.
6: **until** Objective function of Eq. 1 converges.
7: Compute $\mathbf{R}$ in Eq. 7 by quadratic programming.
8: Compute $\mathbf{Y}_t$ given $\{\mathbf{V}_t, \mathbf{R}\}$ by Eq. 8.

The parameter learning and prediction procedures for the proposed TVSE is summarized in Algorithm 1. The time complexity of TVSE consists of solving Eq. 1 and Eq. 7. Typically, optimizing Eq. 1 costs complexity of $\mathcal{O}(\sum_{\pi\in\mathcal{I}}(kmn_\pi^2 T + mn_\pi^2 + m^2 n_\pi)$, where T is the is the number of iterations. For Eq. 1, it requires about $\mathcal{O}(c_s^3)$ time. As $c_s \ll n_\pi$, the time complexity of TVSE is quadratic to the numbers of the source and target instances. We will release our implementation code for public usage.

4 EXPERIMENTS

4.1 Experimental Setup

Datasets. We evaluate our work on four benchmark ZSL datasets: 1) Animals with Attributes (AwA) [13], 2) Caltech UCSD Birds (CUB) [29], 3) aPascal-aYahoo (aPY) [6] and 4) SUN Attribute (SUN) [23], for zero-shot recognition problem. They consist of images belonging to coarse-grained or fine-grained categories in different scopes, such as animals, vehicles and natural scenes. Table 1 summarizes the statistics in each dataset. In our experiments, we utilize the same experimental settings as [34], including the splits of source/target images and seen/unseen classes. Regarding the available semantic representations of classes, we use the provided attributes of each dataset. For AwA and aPY datasets, we directly utilize the provided class-level attribute vector of continuous value. For CUB and SUN datasets which have binary attribute vector image-level, we take the mean of attribute vectors of all images from the same classes to generate class-level attribute vector.

Table 1: General statistics of the four datasets used in the experiments, where "*/*" in columns "Images" and "Classes" stands for the number of source/target images and seen/unseen classes, respectively.

Datasets	Images	Attributes	Classes
AwA	24,295 / 6,180	85	40 / 10
CUB	8,855 / 2,933	312	150 / 50
aPY	12,695 / 2,644	64	20 / 12
SUN	14,140 / 200	102	707 / 10

Features. For all the datasets, we utilize the deep features extracted from popular CNN architecture. It has been shown that the deep features work remarkably better than the low-level features as they lead to good intra-class separation [2]. Specifically, we use the CNN features originally provided in [34], which are extracted from the 4,096 dimensional top-layer hidden unit activations (fc7) in VGG (very-deep-19) [26] network. Similar CNN features have already been used in previous works [2, 34, 35] for zero-shot recognition.

Settings. We use classification accuracy as the evaluation criteria, as it has been widely adopted in the literature. The proposed TVSE involves five parameters: the regularization coefficients λ, μ and γ, the number of feature cluster k, the number of nearest neighbors p. When compared with other approaches, we empirically set $\{\lambda, \mu, \gamma, k, p\}$ to $1, 100, 100, 4, 16$ for TVSE through all the experiments. We report the averaged results over five runs with the optimal parameters chosen on each dataset. In the coming section, we make a detailed analysis on parameter sensitivity which shows that TVSE is able to achieve stable performance with a wide range of parameter values. All the experiments are conducted on a desktop machine which has 4-core 3.3GHz CPUs with 16GB RAM.

4.2 Main Results

4.2.1 Results on Zero-shot Recognition. To make comprehensive comparison for the recognition task on the four datasets, we compare the proposed TVSE against various state-of-the-art alternatives: 1) the inductive methods, i.e., DAP [13], ALE [1], ESZSL [25], SSE-ReLU [34], JSLE [35] and SynC-struct [5]; 2) the transductive methods, i.e., UDA-ZSL [12] and TMV-HLP [9]; 3) the integrated methods, i.e., three extensions based on SP-ZSR [36].

Table 2 shows the overall classification accuracy of the different methods on the four datasets. It can be observed that: 1) The proposed TVSE clearly outperforms the best inductive methods such as JSLE and SynC-struct by a large margin. The main advantage of TVSE compared with the inductive methods is that it can incorporate the data of both seen and unseen classes during the learning procedure, hence overcomes the test-time projection shift between seen and unseen classes. 2) TVSE also significantly outperforms the other transductive methods UDA-ZSL and TMV-HLP, e.g, 10.9% on AwA and 13.4% on CUB. It indicates the superiority of TVSE on exploring the data structures of both seen and unseen classes via modelling their manifold structures. In addition, compared with the projection based method TMV-HLP, TVSE provides a new perspective to learn a more effective latent embedding space under the transductive setting. 3) TVSE achieves competitive recognition performance against the integrated methods based on SP-ZSR. Specifically, compared with TVSE, the performance of these integrated methods depends on the specific inductive method used and the specific dataset. For example, TVSE obtains better accuracy scores than those of ESZSL + SP-ZSR on aPY and AwA, while performs on par with JSLE + SP-ZSR and BL-ZSL + SP-ZSR on these two datasets. In contrast, TVSE performs inferior to ESZSL + SP-ZSR while obtains better results than JSLE + SP-ZSR and BL-ZSL + SP-ZSR on SUN. 4) While these integrated methods boost the already learned recognition models by the inductive methods such as ESZSL, JSLE and BL-ZSL, the proposed TVSE directly learns promising recognition model more effectively. Thus it indicates that TVSE is promising to account for the test-time projection shift problem as the SP-ZSR does.

4.2.2 Results on Zero-shot Retrieval. In the retrieval task, a semantic embedding vector of an unseen class is used as a query to retrieve top matched test images. The performance is measured by mean average precision (mAP). As the retrieval task is not widely evaluated in prior studies, in this work we provide a comprehensive comparison with the state-of-the-art methods including the inductive methods SSE-ReLU, JSLE and SynC-struct; and the integrated methods based on SP-ZSR.

Table 3 summarizes the overall retrieval results in terms of mAP for all datasets. We can see that our TVSE obtains mAP score of 79.8% compared to the result (51.1%) with a large margin from the best inductive method of SynC-struct. It thus again validates the superiority of our TVSE

Table 2: Comparison of classification accuracy (%) for zero-shot recognition task on the four datasets, in the form of mean±standard deviation, and "-" means no result reported. The best results on each dataset are highlighted in bold font.

Method	aPY	AwA	CUB	SUN	Average
DAP [13]	38.2	57.2	-	72.0	
ALE [1]	-	61.9	40.3	-	-
ESZSL [25]	24.22±2.89	75.32±2.28	-	82.10±0.32	-
SSE-ReLU [34]	46.23±0.53	76.33±0.83	30.41±0.20	82.50±1.32	58.91
JSLE [35]	50.35±2.97	79.12±0.53	41.78±0.52	83.83±0.29	63.77
SynC-struct [5]	48.88±0.77	78.57±3.25	50.32±1.14	81.53±0.54	64.82
UDA-ZSL [12]	-	75.6	40.2	-	-
TMV-HLP [9]	-	80.5	47.9	-	-
ESZSL + SP-ZSR [36]	37.5	84.3	-	89.5	-
JSLE + SP-ZSR [36]	62.19±4.65	**92.08±0.14**	**55.34±0.77**	86.12±0.99	73.93
BL-ZSL + SP-ZSR [36]	**69.74±3.47**	92.06±0.18	53.26±1.04	86.01±1.32	**75.27**
TSVE	66.21±1.02	89.28±0.45	53.87±1.96	**88.05±1.46**	74.35

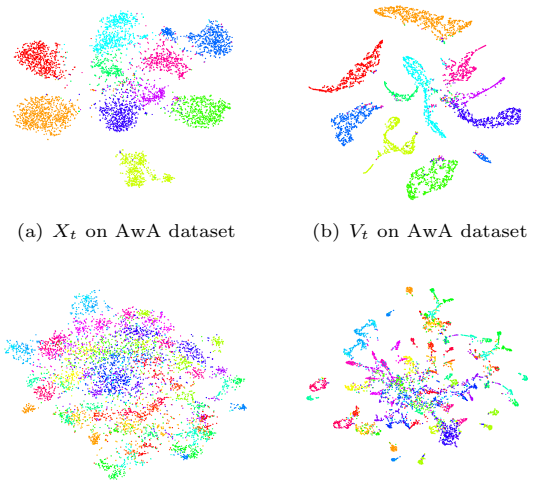
on learning more effective projection under the transductive setting. Furthermore, our TVSE also outperforms the integrated methods ESZSL + SP-ZSR and BL-ZSL + SP-ZSR, and obtains comparative mAP score (79.8%) against that of JSLE + SP-ZSR (80.0%). The competitive performance of TVSE in retrieval is due to the useful manifold structure explored in the test instances, which enhances the learned projection to better match the test instances to the semantic embeddings of unseen classes. Visualized retrieval examples are in the Suppl.

Table 3: Retrieval performance comparison (%) in terms of mAP. The best results on each dataset are highlighted in bold font.

Method	aPY	AwA	CUB	SUN	Ave.
SSE-ReLU [34]	14.1	42.6	3.7	44.6	26.2
JSLE [35]	32.7	66.5	23.9	76.5	49.9
SynC-struct [5]	30.4	65.4	34.3	74.3	51.1
ESZSL + SP-ZSR [36]	37.5	84.3	-	89.5	-
JSLE + SP-ZSR [36]	70.7	**94.0**	**63.2**	92.1	**80.0**
BL-ZSL + SP-ZSR [36]	**74.1**	92.0	58.8	91.2	75.3
TVSE	72.3	92.4	60.9	**93.0**	79.6

4.3 Detailed Analysis on TVSE

4.3.1 Effects of the Latent Embedding Space. To better understand the proposed TVSE on recognizing the target instances of unseen classes, we employ t-SNE [28] to visualize their CNN features (X_t) and mixtures of seen class scores V_t of the visual feature space and the learned latent embedding space, respectively. The four sub-figures in Figure 2 illustrate the visualized distributions of the two spaces on AwA and CUB datasets. On AwA dataset, in Figure 2 (a) the CNN features seem to form separate clusters for the 10 unseen classes with some overlaps, which is useful for recognition. The gaps between each class are much clearer in Figure 2 (b), which indicates TVSE well explores the manifold structures of the target instance under the transductive setting. In contrast, on CUB dataset, in Figure 2 (c) the CNN features are heavily overlapped for the 50 unseen classes due to the much finer granularity in the classes. Nevertheless, TVSE still appears to learn more separable embeddings as seen in Figure 2 (d), where the embeddings appear to be more cluttered with smaller intra-class gaps and overlaps.



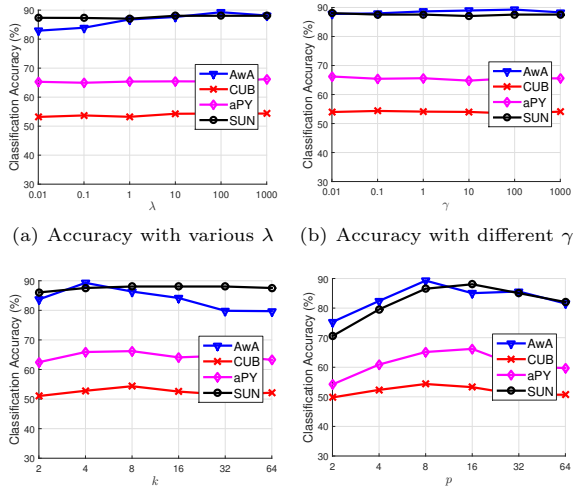
(a) X_t on AwA dataset (b) V_t on AwA dataset

(c) X_t on CUB dataset (d) V_t on CUB dataset

Figure 2: t-SNE visualization comparison of X_t in the visual feature space and V_t in the latent embedding space on the AwA and CUB datasets.

4.3.2 Parameter Sensitivity. To study the parameter sensitivity of TVSE, we tune one parameter at each time while fixing the others to their optimal values. Due to the space limit, we only report the performance variations with four

parameters λ, γ, k and p except for μ, since μ plays the same role as λ. Figure 3 (a) shows the accuracy under varying values of λ. We can observe that TVSE achieves steady performance when λ spans over a wide range in $[10, 1000]$. In contrast, TVSE is not very sensitive to γ as shown in Figure 3 (b), which implies that the proposed scheme for relational mapping of classes in TVSE is robust. Moreover, due to the intrinsic manifold structures in the different dataset, the optimal values of k and p vary depending on the specific dataset. In general, TVSE can obtain considerable performance when $k \in [4, 16]$ and $p \in [16, 32]$.



(a) Accuracy with various λ (b) Accuracy with different γ

(c) Accuracy with different k (d) Accuracy with different p

Figure 3: Parameter sensitivity of TVSE on the four datasets.

4.3.3 Run Time Comparison. In this subsection, we evaluate the efficiency of TVSE in comparison with several baseline methods. We record the training time costs for all methods with their best performance obtained. Table 4 shows the training and test time for all methods on AwA. Firstly, we can observe that the training time cost for different method is various. In particular, the inductive methods JSLE and SynC-struct cost much more time to train the models and to conduct prediction on test-data than SSE-ReLU due to the heavy load of computing the structured loss term of pairwise instances during optimizing and prediction. Thus it implies that they may not be suitable for large-scale data. Secondly, it can be learned that the transductive method TMV-HLP and the integrated methods based on SP-ZSR generally need more time for joint learning and prediction than the inductive methods. Though our TVSE also forms the learning and prediction procedures jointly, it requires much less time than TMV-HLP and the integrated methods based on SP-ZSR. Note that it even runs faster than the inductive method SynC-struct. As TVSE needs to calculate the affine matrix (manifold information) of the test instances and iteratively updates the prediction result, it is essential for TVSE to achieve superior prediction performance to TMV-HLP and SynC-struct. Therefore, TVSE can learns the prediction model with acceptable running time and comparative performance.

Table 4: Comparison of the running time (in seconds) on AwA.

Method	Training	Test
SSE-ReLU [34]	114.1	0.7
JSLE [35]	332.7	6.9
SynC-struct [5]	500.5	9.4
TMV-HLP [9]	745.2	
ESZSL + SP-ZSR [36]	619.5	
JSLE + SP-ZSR [36]	869.3	
TVSE	376.4	

5 CONCLUSION

In this paper, we proposed a novel transductive visual-semantic embedding (TVSE) framework for zero-shot learning. Different from most existing ZSL approaches, TVSE resolves the inherent projection shift problem by learning a latent embedding space using labeled source instances and unlabeled target instances under the transductive setting. We employed collective matrix factorization with graph regularization to incorporate the manifold structures of instances in the latent embedding space. Then we developed a scheme for TVSE to construct the relational mapping between seen and unseen classes, which maps the representations of the target instances in the latent embedding space to their predictions of unseen classes. We compared TVSE against several state-of-the-art approaches over four widely used benchmark datasets, and the extensive evaluations validate the advantages and show the promising capability of TVSE over these alternatives on the zero-shot recognition and retrieval tasks.

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